

Performance Modelling for Computer Systems

Assignment Solutions

Q1: We can denote the state of the system as the number of jobs in the system. For all states n such that $n \leq 2K - 1$, it is going to state $n + 1$ with rate λ . States with $1 \leq n \leq K - 1$ go to state $n - 1$ with rate $n\mu$ and all other states go to $n - 1$ with rate $K\mu$.

The Q matrix is

$$\begin{bmatrix} -\lambda & \lambda & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ \mu & -(\mu + \lambda) & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 2\mu & -(2\mu + \lambda) & \lambda & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 0 & 3\mu & -(3\mu + \lambda) & \lambda & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & K\mu & -(K\mu + \lambda) & \lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & K\mu & -(K\mu + \lambda) & \lambda \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & K\mu & -K\mu \end{bmatrix}$$

The stationary distribution is

$$\pi_i = \left(\frac{\lambda}{\mu}\right)^i \frac{1}{i!} \pi_0 \quad \text{for } i \leq K$$

$$\pi_i = \left(\frac{\lambda}{\mu}\right)^i \frac{1}{K!K^{i-K}} \pi_0 \quad \text{for } K < i \leq 2K$$

The blocking probability is π_{2K} .

Note: The question of whether the stationary distribution has a closed form is ambiguous and won't be considered while evaluating your answers.

Q2: Refer to Figure 1.

Q3: There are 6 states for the system which are as follows (Refer to Figure 2):

- (a) Both servers are empty
- (b) Server 1 has job C_1 .
- (c) Server 2 has job C_2 .
- (d) Server 1 has job C_1 and server 2 has job C_2 .
- (e) Server 1 has job C_2 .
- (f) Server 1 and server 2 both have C_2 .

The Q matrix for the process is as follows

$$\begin{bmatrix} -(\lambda_1 + \lambda_2) & \lambda_1 & \lambda_2 & 0 & 0 & 0 \\ \mu_1 & -(\mu_1 + \lambda_2) & 0 & \lambda_2 & 0 & 0 \\ \mu_2 & 0 & -(\mu_2 + \lambda_1 + \lambda_2) & \lambda_1 & 0 & \lambda_2 \\ 0 & \mu_2 & \mu_1 & -(\mu_1 + \mu_2) & 0 & 0 \\ \mu_2 & 0 & 0 & 0 & -(\mu_2 + \lambda_2) & \lambda_2 \\ 0 & 0 & \mu_2 & 0 & \mu_2 & -(2\mu_2) \end{bmatrix}$$

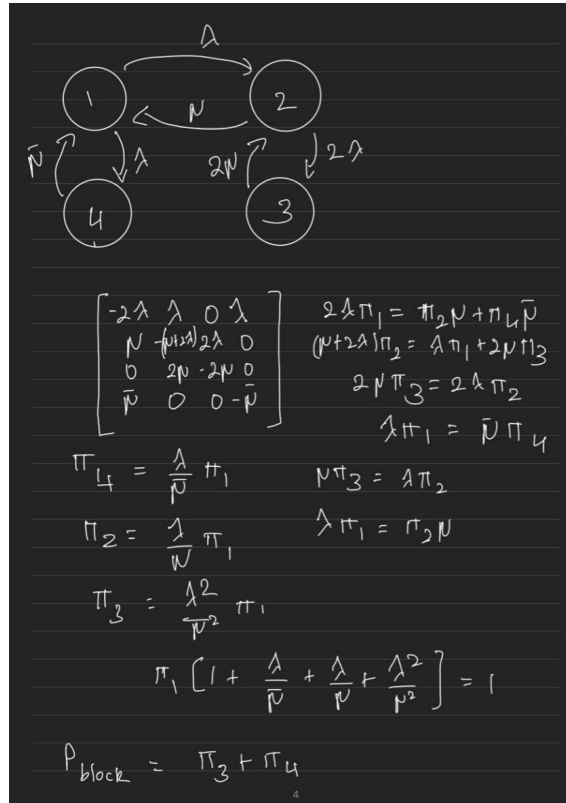


Figure 1: Q2 Solution

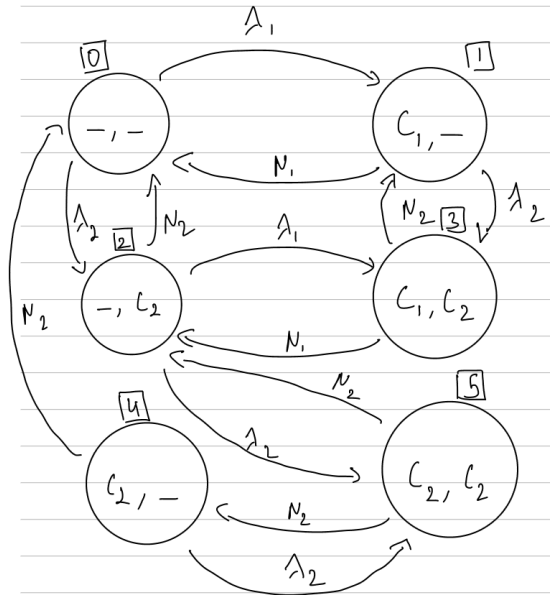


Figure 2: Markov Chain for Q3

This can be solved using the method of partial balance/local balance taught in class(Refer to section 17.4 from Mor Harchol-Balter)

Q4: (a) Check Q2 solution.

(b) Let the stationary distribution be $\pi = [\pi_0 \cdots \pi_5]$. Then

$$\begin{aligned} P(\text{Blocking}) &= P(\text{Blocking}|C_1)P(C_1) + P(\text{Blocking}|C_2)P(C_2) \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2}(1 - \pi_0 - \pi_2) + \frac{\lambda_2}{\lambda_1 + \lambda_2}(\pi_3 + \pi_5) \end{aligned}$$

Q5: Refer to sections 13.1, 13.2, 14.3, 14.4 from Mor Harchol-Balter.

Q6: Let the time at which the doors are closed be given by T . Then the expected number of arrivals is λT . Thus, the expected number of people who come to class before doors are closed is given by

$$\int_0^{15} \lambda t P(T = t) dt$$

Since T is given by a uniform distribution with range from 0 to 15. We can rewrite this as

$$\begin{aligned} \int_0^{15} \lambda t \frac{1}{15} dt &= \frac{1}{15} \int_0^{15} \lambda t dt \\ &= \frac{1}{15} \frac{15^2}{2} \lambda \\ &= 7.5\lambda \end{aligned}$$

Q7: Clearly $\mathbb{P}(N = 0) = 1 - f_{ij}$, while, by conditioning on the time of the n th visit to j , we have that $\mathbb{P}(N \geq n + 1 \mid N \geq n) = f_{jj}$ for $n \geq 1$, whence the answer is immediate. Now $\mathbb{P}(N = \infty) = 1 - \sum_{n=0}^{\infty} \mathbb{P}(N = n)$ which equals 1 if and only if $f_{ij} = f_{jj} = 1$.

Q8: Refer to figure 3.

Q9: Refer to figure 4.

- ⑧
- State 0 (0,0) both S_1, S_2 idle
 State 1 (1,0) S_1 busy, S_2 idle
 State 2 (0,1) S_1 idle, S_2 busy
 State 3 (1,1) both S_1, S_2 busy

$$Q = \begin{bmatrix} -1 & 1 & 0 & 0 \\ \mu_1 & -(\mu_1+1) & 0 & 1 \\ \mu_2 & 0 & -(\mu_2+1) & 1 \\ 0 & \mu_2 & \mu_1 & -(\mu_1+\mu_2) \end{bmatrix}$$

$$\pi Q = 0$$

$$\sum \pi_i = 1$$

$$1\pi_0 = \mu_1\pi_1 + \mu_2\pi_2 \quad \text{--- ①}$$

$$(\mu_1+1)\pi_1 + 1\pi_0 + \mu_2\pi_3 \quad \text{--- ②}$$

$$(\mu_2+1)\pi_2 = \mu_1\pi_3 \quad \text{--- ③}$$

$$(\mu_1+\mu_2)\pi_3 = 1(\pi_1+\pi_2) \quad \text{--- ④}$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \quad \text{--- ⑤}$$

Blocking probability (The probability that both servers S_1 and S_2 are busy = State 3)

$$P_{\text{block}} = \pi_3 \quad \text{--- Solve for } \pi_3$$

Figure 3: Q8 solution

9) generator matrix $Q = \frac{d}{dt} P(t) \Big|_{t=0}$

$$Q = \begin{bmatrix} -1 & 1/2 & 1/2 \\ 1/6 & -1/3 & 1/6 \\ 1/6 & 1/12 & -1/4 \end{bmatrix}$$

→ Solve for stationary distribution

$$\pi Q = 0 \quad \text{and} \quad \pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_1 + \pi_2 + \pi_3 = 1 \quad \text{--- (1)}$$

$$-\pi_1 + \frac{1}{6}\pi_2 + \frac{1}{6}\pi_3 = 0 \quad \text{--- (2)}$$

$$\frac{1}{2}\pi_1 = \frac{1}{3}\pi_2 + \frac{1}{12}\pi_3 = 0 \quad \text{--- (3)}$$

$$\frac{1}{2}\pi_1 + \frac{1}{6}\pi_2 - \frac{1}{4}\pi_3 = 0 \quad \text{--- (4)}$$

$$\therefore \pi = \left(\frac{5}{35}, \frac{12}{35}, \frac{18}{35} \right)$$

→ Steady state convergence

$$P(t \rightarrow \infty) = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/3 & 1/3 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix}$$

Since the rows of $P(t)$ do not converge to the same vector the system does not reach a Steady state

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Figure 4: Q9 solution